ENTERPRISE SALES ANALYTICS DATA PLATFORM (ESADP)

Overall Project Documentation — Sprints 0 to 3

UC19: Building a Unified Sales Performance & Revenue Analytics Platform

Enterprise Data Engineering Capstone Project

Submitted by

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Client: ABC Global Retail Ltd.
Methodology: Agile Scrum (Sprints 0, 1, 2 and 3)
Technology: Pentaho Data Integration, PostgreSQL, Python/Pandas, SQL, Power BI and Git

Project Repository: https://github.com/kusumasreed2026-vit/ESADP

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1. Project Charter and Business Requirements

1.1 Project Charter

Field	Content
Project Name	Enterprise Sales Analytics Data Platform (ESADP)
Sponsor	VP of Sales & Revenue, ABC Global Retail Ltd. (fictional)
Goal	Integrate sales, customer, product, seller, pricing, marketing and inventory data into a governed PostgreSQL warehouse with Power BI reporting.
Duration	4 Sprints (Sprint 0 initiation plus Sprints 1 to 3)
Methodology	Agile Scrum
Team	Hitashri M (26MML1024), Kusuma Sree (26MML1021), Krishnaveni M (26MML1008)
Success Criteria	Reproducible source-to-warehouse pipeline, documented mappings and lineage, validation evidence and executive analytics dashboard.
Data Basis	Olist Brazilian E-Commerce data, Olist Marketing Funnel data and a self-generated inventory dataset.

1.2 Business Context

ABC Global Retail Ltd. represents a retail and e-commerce organization whose sales, customer, product, seller, payment, marketing and inventory information originates in independent systems. Point of Sale, Sales Order Management, CRM, ERP, Product Catalog, Pricing, Inventory, Marketing and Dealer Management data were held separately, so no single view of sales performance, revenue trend, inventory position or customer behaviour existed. ESADP consolidated selected datasets into a PostgreSQL analytical store and presented the resulting measures through Power BI.

1.3 Stakeholders and Business Needs

Stakeholder	Need
VP, Sales & Revenue	Unified view of sales performance and revenue trends.
Regional Sales Manager	Compare sales and delivery outcomes across regions.
CRM / Customer Insights Lead	Understand customer purchase and review behaviour.
Merchandising / Product Lead	Identify high-performing product categories.
Dealer / Partner Manager	Compare seller contribution to sales.
Inventory Planner	Compare inventory levels with demand forecasts.
Marketing Analyst	Review lead sources and closed deals.
Data Engineering / Governance	Maintain ETL, definitions, lineage and validation evidence.

1.4 Business Objectives and How They Were Met

The objectives set in Sprint 0 were consolidating selected sources, automating ingestion, documenting data definitions and lineage, validating warehouse outputs, and delivering two Power BI pages. The table below shows how each objective was addressed.

Objective	How ESADP Delivered It	Reference
Consolidate sales order, customer, product, seller, pricing, marketing and inventory data into one warehouse	Source data was ingested through Pentaho into PostgreSQL staging and conformed into a Gold star schema.	Sections 2, 5, 8
A single, trusted view of revenue and regional performance	The Executive Overview page presents KPI cards, revenue by state and the monthly revenue trend.	Sections 9.1, 10.1
Product and category performance without manual joins	dim_product filters fact_sales; top categories are ranked by revenue.	Sections 8, 9.1
Compare seller performance	dim_seller filters fact_sales; the Top 10 Sellers visual ranks sellers by sales.	Sections 8, 9.2
Stock position alongside demand	fact_inventory feeds the inventory versus demand forecast visual.	Sections 8, 9.2
Lead-to-deal conversion alongside sales outcomes	fact_marketing_leads and fact_closed_deals feed the leads-by-origin and closed-deals-by-segment visuals.	Sections 8, 9.2
Every dashboard number traceable to its source	Source-to-target mapping, metadata repository, lineage artifacts, validation SQL and audit logs were produced.	Section 7

2. Source Inventory and Data Dictionary

2.1 Source Inventory

Ten enterprise systems were represented using real, publicly available data, with the self-generated inventory dataset completing the set. Each source was preserved in, or converted to, the format shown below for ingestion.

#	Source Area	Dataset / Source	Format in Ingestion Plan
1	Sales Order Management	olist_orders_dataset.csv	CSV native
2	Point of Sale	olist_order_items_dataset.csv	SQL / database representation
3	CRM	olist_customers_dataset.csv	JSON converted
4	E-Commerce Platform	Shared transactional core; not duplicated	JSON representation documented
5	ERP / Payments	olist_order_payments_dataset.csv	XML converted
6	Dealer Management	olist_sellers_dataset.csv	XML converted
7	Product Catalog	olist_products_dataset.csv + category translation	Excel converted
8	Pricing Management	price and freight_value from order items	Excel converted
9	Inventory Management	Self-generated inventory data	SQL / database source
10	Marketing Campaign	Qualified leads and closed deals datasets	CSV native

2.2 Data Dictionary

The data dictionary records source system, field name, data type, business meaning and a sample value for every staged field, and is aligned with the actual staging and warehouse schemas. The excerpt below shows a representative field from each source area.

Source Area	Field	Data Type	Meaning	Sample Value
Sales Order Management	order_id	VARCHAR	Identifier for a customer order.	e481f51...9dc
Sales Order Management	order_purchase_timestamp	TIMESTAMP	Timestamp when an order was placed.	2018-01-25 14:22:31
CRM	customer_id	VARCHAR	Identifier for a customer record.	3f5c8a2...e17
Product Catalog	product_category_name	VARCHAR	Product category.	beleza_saude
Pricing Management	price	NUMERIC (10,2)	Price of an order item.	89.90
Dealer Management	seller_id	VARCHAR	Marketplace seller identifier.	3442f80...b23
Inventory Management	stock_quantity	INTEGER	Recorded stock quantity.	145
Marketing Campaign	mql_id	VARCHAR	Marketing-qualified lead identifier.	dc4802a...f6b

2.3 Governance Note

Inventory product IDs are not assumed to match Olist product IDs. Inventory was therefore kept separate from dim_product, and no join between the two is made unless an explicit, validated crosswalk exists.

3. Solution Architecture and Technology Stack

3.1 End-to-End Architecture

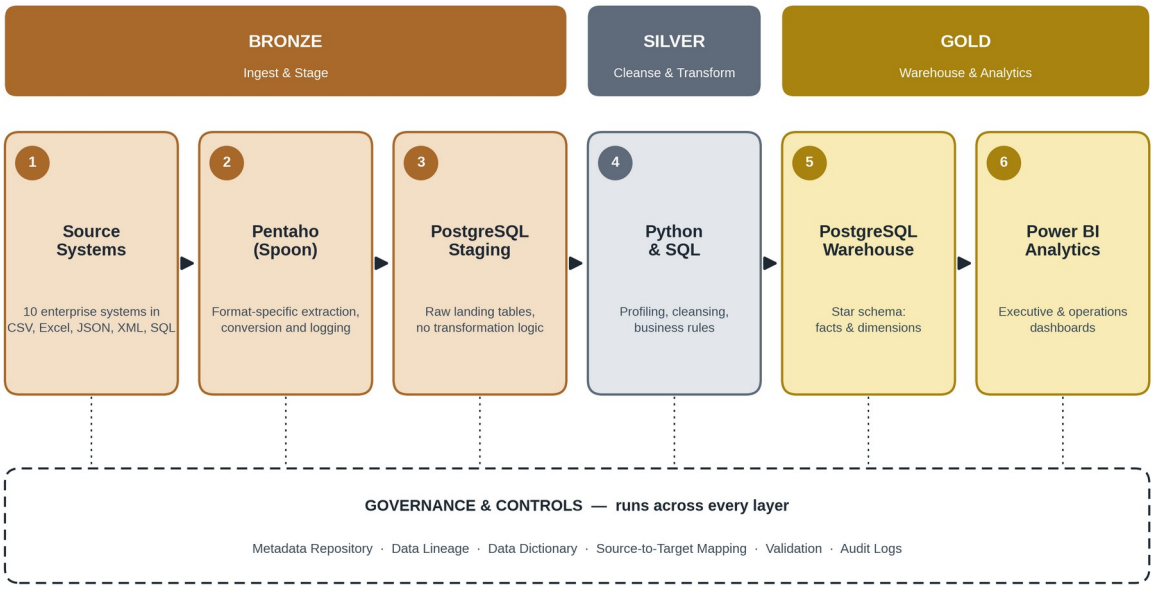


Figure 1. ESADP End-to-End Data Architecture (Medallion Design: Bronze – Silver – Gold)

The platform follows a medallion architecture. The Bronze zone extracts, converts, routes and lands source records in PostgreSQL staging exactly as received. The Silver zone profiles, validates, standardizes and transforms that data. The Gold zone holds the analytics-ready PostgreSQL star schema of fact and dimension tables and the Power BI report pages. Metadata, lineage, glossary, audit logs and validation operate as cross-cutting governance controls across all three zones.

3.2 End-to-End Process Flow

1. Real datasets (Olist e-commerce, Olist marketing funnel and the self-generated inventory dataset) were identified and mapped to the ten enterprise systems.
2. Each source was exported or converted into its designated format: CSV, Excel, JSON, XML or SQL.
3. Pentaho transformations ingested each format and loaded the data, unmodified, into matching PostgreSQL Bronze staging tables.
4. Python scripts cleansed and validated the staged data for duplicates, missing values and inconsistent categories in the Silver zone.
5. A star schema was designed, and the PostgreSQL Gold warehouse was populated from the Silver-layer data.
6. Governance documentation — data dictionary, lineage, source-to-target mapping and glossary — was maintained alongside every layer.
7. Power BI connected to the Gold layer and delivered the executive and operational dashboards.
8. The full pipeline was orchestrated through Pentaho jobs, validated through reconciliation checks and released through a tagged Git commit.

3.3 Role of PostgreSQL

PostgreSQL served two distinct roles. First, it was the staging layer: a raw landing zone where each source system landed in its own staging table, structurally unmodified from source, keeping ingestion (Sprint 1) and cleansing (Sprint 2) as separate, auditable steps. Second, it was the warehouse layer: the analytics-ready store, held in a separate schema of the same database, containing the star schema that Power BI connects to directly.

Using one relational engine for both roles kept the scope appropriate for the project while still requiring genuine schema design, indexing and SQL query-writing. In short, PostgreSQL acted as both the inbox (staging) and the filing system (warehouse): everything upstream of it is format conversion, and everything downstream of it is reporting.

3.4 Layer Responsibilities

Layer	Technology	Responsibility
Source	CSV, Excel, JSON, XML and database	Provide raw records for represented enterprise systems.
Bronze	Pentaho and PostgreSQL staging	Extract, convert, route and land records.
Silver	Python/Pandas and SQL	Profile, validate, standardize and transform data.
Gold	PostgreSQL warehouse	Store analytics-ready facts and dimensions.
Analytics	Power BI	Present KPI cards, trends and business visuals.

Layer	Technology	Responsibility
Governance	SQL, documentation and Git	Record mappings, lineage, definitions, validation and version history.

3.5 Technology Stack

Technology	Use
Pentaho Data Integration (Spoon)	Format-specific transformations and job orchestration.
PostgreSQL / pgAdmin	Staging, warehouse, metadata and validation queries.
Python / Pandas	Profiling, data-quality inspection and cleaning.
Power BI Desktop	Two-page business dashboard.
Git	Version control for ETL, SQL, scripts and documentation.

3.6 Key Design Decisions

Decision	Rationale
Medallion (Bronze / Silver / Gold) architecture	Separates raw landing, quality processing and analytics-ready storage so that each step is auditable.
One PostgreSQL database with separate staging and warehouse schemas	Keeps the platform focused while still exercising schema design, indexing and SQL.
Ingestion kept separate from cleansing	Raw data stays untouched in staging, so quality rules can be reviewed and rerun against the original.
Inventory kept separate from dim_product	No validated crosswalk exists between inventory product IDs and Olist product IDs.
Purchase date as the active date relationship	fact_sales has several date attributes; purchase date drives the dashboard and the others are activated only in measures.
Facts held at native grain	Order-item-level sales support product and seller analysis; distinct counts are used where order-level measures are required.
Governance as a parallel layer	Mapping, lineage, metadata and validation cover every layer so each dashboard number can be traced to its source.

4. Repository Structure and Sprint Summary

4.1 Repository Structure

```
ESADP/
├── docs/
├── config/
├── datasets/
│   ├── bronze/
│   │   ├── ingestion/
│   │   └── staging/
│   ├── silver/
│   │   ├── cleansing/
│   │   ├── validation/
│   │   └── transformations/
│   └── gold/
│       ├── warehouse/
│       ├── datamarts/
│       └── analytics/
├── pentaho/
│   ├── transformations/
│   └── jobs/
├── python/
├── sql/
├── metadata/
├── lineage/
├── dashboards/
├── deployment/
├── tests/
└── README.md
```

The repository was initialised and maintained collaboratively by the team throughout the project, with all three members contributing commits ahead of each submission milestone.

4.2 Folder Purposes

Folder	Purpose
docs/	Project and sprint documentation.
config/	Configuration for connections and execution.
datasets/bronze/ingestion, staging	Source files and staging objects for the raw landing zone.
datasets/silver/cleansing, validation, transformations	Cleansing outputs, data-quality validation and business-rule transformations.
datasets/gold/warehouse, datamarts, analytics	Star-schema warehouse objects, subject-area views and analytics outputs.
pentaho/transformations, jobs	Pentaho .ktr transformations and .kjb orchestration jobs.
python/	Profiling and cleaning scripts.
sql/	DDL, transformations and validation queries.
metadata/	Metadata definitions and scripts.
lineage/	End-to-end and column-level lineage.
dashboards/	Power BI report and exports.

Folder	Purpose
deployment/	Release and deployment material.
tests/	Validation scripts and evidence.
README.md	Setup, execution and file locations.

4.3 Sprint Summary

Sprint	Focus	Key Outputs
0	Project initiation and architecture	Charter, BRD, source inventory, architecture and backlog.
1	Source conversion, ingestion and staging	Converted sources, Pentaho jobs, staging tables and initial data dictionary.
2	Profiling, quality and warehouse preparation	Profiling reports, quality findings, cleaning rules and warehouse outputs.
3	Governance, orchestration, validation and BI	Mapping, metadata, lineage, validation, audit evidence and dashboard.

4.4 Sprint 0 and Sprint 1 Backlog

ID	User Story	Sprint	Status
PB-01	Scaffold the Git repository using the medallion folder structure so all members commit consistently.	0	Completed
PB-02	Draft and finalize the Business Requirement Document to lock project scope before development.	0	Completed
PB-03	Design the solution architecture, covering every medallion layer and the role of PostgreSQL.	0	Completed
PB-04	Build the sprint backlog to plan and track work across all sprints.	0	Completed
PB-05	Export each of the 10 source systems into its designated format (CSV, Excel, JSON, XML, SQL).	1	Completed
PB-06	Build Pentaho transformations converting each format and loading it into PostgreSQL Bronze staging tables.	1	Completed
PB-07	Compile a complete data dictionary covering every staged field.	1	Completed
PB-08	Implement logging and exception handling on every ingestion pipeline.	1	Completed
PB-09	Maintain incremental Git commit history showing ingestion progress.	1	Completed

4.5 Traceability Principle

Every major deliverable is linked to evidence: ETL files and run logs for ingestion, profiling and before/after reports for data quality, schema and row counts for the warehouse, and mapping, lineage and validation artifacts for governance. Every design decision in later sprints traces back to a scope or architecture decision documented in Sprint 0.

5. Sprint 1 — Source Conversion, Ingestion and Staging

Sprint 1 established the ingestion path from source datasets to PostgreSQL staging. Pentaho transformations read each agreed source format and loaded its corresponding staging target.

5.1 Ingestion Flow

1. Each source file or table and its source-system area was identified.
2. Data was converted into the agreed format where required.
3. Pentaho input steps and field metadata were configured.
4. Records were loaded into the matching PostgreSQL staging table.
5. Row counts and execution logs were checked, and connection, schema and mapping errors were resolved.

5.2 Source-to-Format Ingestion Map

Source Area	Format	Pentaho Ingestion Approach
Sales Order Management	CSV	CSV file input loaded to the orders staging table.
Point of Sale	SQL	Table input from the database representation loaded to staging.
CRM	JSON	JSON input loaded to the customers staging table.
E-Commerce Platform	JSON	Shared transactional core documented, not re-ingested.
ERP / Payments	XML	XML input loaded to the payments staging table.
Dealer Management	XML	XML input loaded to the sellers staging table.
Product Catalog	Excel	Excel input loaded to the products staging table, including category translation.
Pricing Management	Excel	Excel input loaded to the pricing staging table.
Inventory Management	SQL	Table input from the inventory database source loaded to staging.
Marketing Campaign	CSV	CSV file input for qualified leads and closed deals loaded to staging.

5.3 Staging Principles

Principle	Application
Separate ingestion from cleansing	Source landing was kept distinct from quality processing.
Explicit schema mapping	Source and target fields agree in names, types and mapping.
Reproducible jobs	Pentaho jobs execute the transformations in order.
Failure visibility	Logs and error messages were retained.
Source traceability	Source identifiers and file names were preserved where useful.

5.4 Logging and Exception Handling

Logging and exception handling were implemented on every ingestion pipeline. Execution logs capture the run status of each transformation, and exceptions raised by connection, schema or mapping problems were recorded and resolved before the staging load was accepted.

5.5 Evidence Retained

The working .ktr/.kjb files, the source-to-staging table list, successful run screenshots, and PostgreSQL checks showing the intended staging targets and row counts were retained as Sprint 1 evidence.

6. Sprint 2 — Profiling, Quality and Warehouse Preparation

Sprint 2 examined source and staged data, identified quality issues, applied documented cleaning and standardization rules, and prepared analytics-ready warehouse structures. Profiling captured the pre-conversion baseline where required.

6.1 Data Profiling Report Contents

Item	What It Records
Row count	Number of records in the inspected file or table.
Column count and names	Number of fields and their exact names.
Data types	Detected types before and, where relevant, after conversion.
Missing values	Null/blank count and missing percentage per column.
Duplicate records	Duplicate rows under the selected criteria.
Unique values	Distinct counts and examples where practical.
Basic statistics	Numeric summaries, ranges, date ranges and category distributions.
Quality observations	Invalid formats, inconsistent labels, outliers or suspicious values.

6.2 Cleaning and Quality Rules

Concern	Rule Documentation and Treatment
Missing values	Affected fields, counts, percentages and the handling rule were documented.
Duplicates	Duplicate criteria, counts and the deduplication approach were documented.
Inconsistent categories/formats	A standardization mapping and canonical values were defined.
Type conversion failures	Affected fields, invalid examples and their disposition were recorded.
Key integrity	Null and unmatched key checks were run.
Business-rule anomalies	Each rule, the affected records and the action taken were recorded.

6.3 Warehouse Preparation

Cleansed Silver-layer data was used to populate the Gold star schema: date, customer, product and seller dimensions, and fact tables for sales, payments, reviews, inventory, marketing leads and closed deals. The SQL and transformation files that populate the final staging and warehouse structures were versioned in the repository.

6.4 Sprint 2 Evidence

One profiling report per file or table (or an indexed combined report), the cleaning rules and exception handling, before-and-after quality comparisons, and the SQL and transformation files used to populate the final structures were retained as Sprint 2 evidence.

7. Sprint 3 — Governance, Lineage, Orchestration and Validation

Sprint 3 consolidated the warehouse and reporting work into a traceable and testable deliverable. The focus was source-to-target documentation, metadata, lineage, orchestration, validation and evidence that Power BI uses the intended warehouse model.

7.1 Source-to-Target Mapping

The mapping workbook records source tables and columns, target schema/table/columns, transformation rules, expected data types, key roles and null/default handling. It agrees with the final Pentaho transformations and PostgreSQL schemas.

7.2 Metadata Repository

The metadata repository stores technical metadata about pipeline assets. Its SQL and run guide were executed, and the expected metadata records were queried to confirm that the repository is populated and verified.

7.3 Lineage Artifacts

Artifact	Purpose	Final Review
End-to-end lineage diagram	Shows source → staging → warehouse → dashboard.	Matches final source areas, jobs, schemas and reports.
Column-level lineage	Traces output columns to source fields and transformations.	Compared with the mapping workbook and SQL.
Source-to-target workbook	Detailed field-level mappings.	Target names, keys, types and rules checked.

7.4 Orchestration and Audit

Pentaho jobs coordinate the transformations in a defined order, and the complete required sequence ran to completion. Audit logs capture execution status, timestamps, row counts and errors for each run.

7.5 Validation and Reconciliation

Category	Check Performed
Completeness	Source and target counts were compared, accounting for documented filters.
Uniqueness	Business keys were checked for duplicates.
Null handling	Required identifiers, dates and measures were checked for nulls.
Referential integrity	Fact keys were checked against the expected dimension keys.
Business reconciliation	Selected totals were compared with independently calculated source baselines.
Dashboard consistency	Visual measures and aggregations were confirmed and sample values reconciled to SQL.

7.6 Sprint 3 Artifacts

Artifact	Status
ESADP_Sprint3_Source_to_Target_Mapping.xlsx	Reconciled with final transformations and warehouse.
03_metadata_repository.sql and run guide	Executed; repository objects and records verified.
03_validation.sql and validation workbook	Rerun after the final load; measured results recorded.
End-to-end lineage PNG	Confirmed to reflect the final flow.
Column-level lineage Markdown	Checked against the final mapping.

8. PostgreSQL Warehouse Model

The Gold warehouse includes date, customer, product and seller dimensions, plus fact tables for sales, payments, reviews, inventory, marketing leads and closed deals. The PostgreSQL definitions are the authoritative schema.

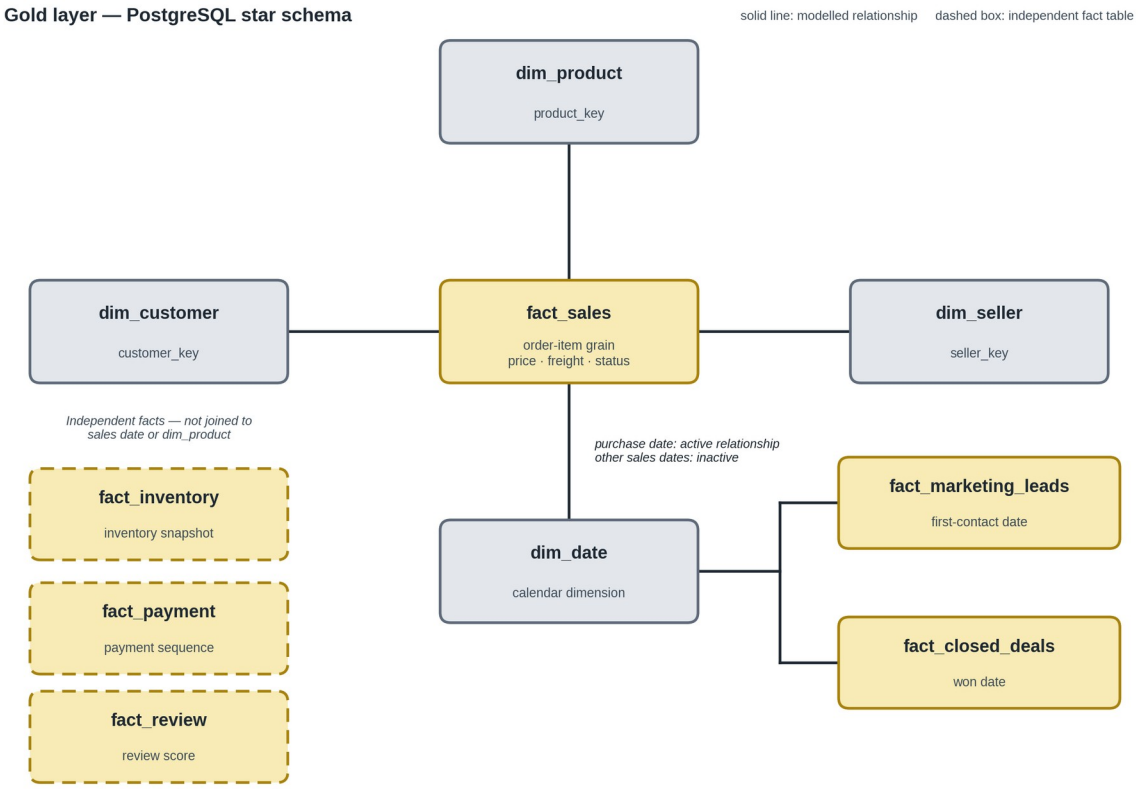


Figure 2. Gold-layer star schema and Power BI relationships

Table	Role and Reporting Notes
dim_date	Calendar dimension; purchase-date relationship to sales is active.
dim_customer	Customer attributes; related to fact_sales by customer_key.
dim_product	Product/category attributes; related to fact_sales by product_key.
dim_seller	Seller attributes; related to fact_sales by seller_key.
fact_sales	Order-item-level sales including order, product, seller, dates, status, price and freight.
fact_payment	Payment records per order/payment sequence.
fact_review	Customer review records and scores.
fact_inventory	Independent inventory snapshot source; not joined to dim_product without mapping.
fact_marketing_leads	Marketing-qualified leads, origin and first-contact date.
fact_closed_deals	Closed-deal records, business segment and won date.

8.1 Fact Table Grain

Fact Table	Grain (one row per)
fact_sales	Order item.
fact_payment	Order and payment sequence.
fact_review	Customer review record.
fact_inventory	Inventory snapshot record.
fact_marketing_leads	Marketing-qualified lead.
fact_closed_deals	Closed deal.

8.2 Metric Definitions

Metric	Definition
Total Sales Revenue	Sum of price plus freight_value in fact_sales.
Total Orders	Distinct count of order_id.
Average Order Value	Total Sales Revenue divided by Total Orders.
Total Customers	Distinct count of customer_key represented in fact_sales.
Delivered Orders	Distinct orders whose status is delivered.
On-Time Delivery Rate	On-time delivered orders divided by delivered orders, using available delivery and estimated dates.
Total Closed Deals	Count of rows in fact_closed_deals.

8.3 Power BI Relationship Notes

- dim_customer, dim_product and dim_seller filter fact_sales through their keys.
- dim_date has multiple sales-date relationships; purchase date is active and other sales-date relationships are inactive unless activated in a measure.
- dim_date is related to fact_marketing_leads and fact_closed_deals by the relevant date keys.
- fact_payment and fact_review are disconnected from the sales date dimension in the current model.
- fact_inventory remains separate from dim_product until a validated crosswalk exists.

9. Power BI Dashboard and Business Interpretation

The submitted dashboard export contains two pages with a navy title band, teal/blue visuals and white backgrounds.

9.1 ESADP | Executive Overview

The first page includes a date-range slicer, sales revenue by state, order volume by month, monthly sales revenue trend, top product categories by revenue, order status distribution, and KPI cards for average order value, total sales revenue, on-time delivery rate and total orders.



Figure 3. Executive Overview page from the supplied dashboard export (page 1).

9.2 ESADP | Business Performance & Operations

The second page includes marketing leads by origin, closed deals by business segment, installments by payment method, top sellers by sales, customer review score distribution, and inventory levels versus demand forecast.

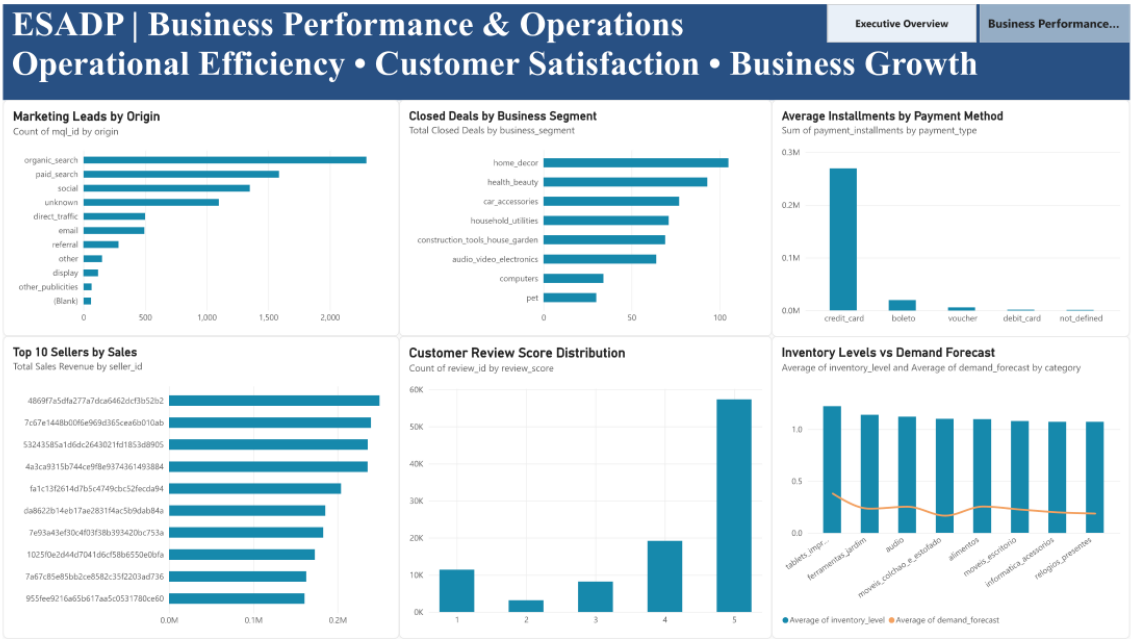


Figure 4. Business Performance & Operations page from the supplied dashboard export (page 2).

9.3 Dashboard Validation

Visual	Business Question	Validation Outcome
KPI cards	What are the headline sales indicators?	Headline indicators reconciled to warehouse SQL; average order value equals total sales revenue divided by total orders.
Sales revenue by state	Where is recorded revenue generated?	Geographic categorization of the state field reviewed against revenue by state.
Monthly trends	How do revenue and order volume change?	Month-level axis and purchase-date filtering confirmed.
Order status distribution	How are order records distributed by status?	Row count versus distinct orders reconciled to Total Orders.
Marketing leads by origin	Which origins contribute the most leads?	Lead counts confirmed against mql_id.
Closed deals by segment	Which segments have the most closed deals?	Aggregation confirmed as a deal count by segment.
Installments by payment method	What is the typical installment count?	Aggregation and subtitle of the installments measure reviewed.
Top sellers by sales	Which sellers contribute most revenue?	Seller labels retain traceability to seller_id.
Review score distribution	How are review scores distributed?	Review-record count and score order 1–5 confirmed.

Visual	Business Question	Validation Outcome
Inventory vs demand forecast	How do inventory and forecast compare?	Inventory kept separate from Olist products, with no unvalidated mapping.

9.4 Stakeholder Coverage

Stakeholder	Dashboard Visuals Serving the Need	Page
VP, Sales & Revenue	KPI cards, monthly revenue trend, revenue by state	Executive Overview
Regional Sales Manager	Revenue by state, order status, on-time delivery rate	Executive Overview
CRM / Customer Insights Lead	Customer review score distribution, installments by payment method	Business Performance
Merchandising / Product Lead	Top 8 product categories by sales revenue	Executive Overview
Dealer / Partner Manager	Top 10 sellers by sales	Business Performance
Inventory Planner	Inventory levels versus demand forecast	Business Performance
Marketing Analyst	Marketing leads by origin, closed deals by business segment	Business Performance

10. Dashboard Insights and Inferences

This section interprets the two dashboard pages. Figures are read from the dashboard export and are approximate unless a visual labels them exactly. The inferences describe what the data suggests; they are not causal findings.

10.1 Executive Overview

Area	Inference
Business scale	The platform reports roughly 99K distinct orders, total sales revenue of about 15.84M and an average order value of 160.58. The three cards are mutually consistent: revenue divided by orders reproduces the average order value.
Order fulfilment	About 97.8% of order records are delivered, with about 1.05% still shipped and the remaining statuses negligible. The on-time delivery rate of 93.22% implies that roughly 7% of delivered orders arrived after the estimated date.
Order volume and revenue	Monthly order volume ranges from about 8K to 11K between January and August and falls to about 4K–7K from September to December. The monthly revenue trend moves with order volume, peaking at roughly 1.7–1.8M around May and August and dipping to about 0.7M in September. Revenue therefore tracks the number of orders rather than basket value.
Product categories	No single category dominates. beleza_saude (health and beauty) leads at about 1.45M, followed by relogios_presentes (watches and gifts), cama_mesa_banho (bed, bath and table) and esporte_lazer (sports and leisure), each above about 1.1M. The top eight categories span personal care, gifts, home goods and leisure.
Geography	Revenue is concentrated in a limited set of Brazilian states rather than spread evenly, so regional performance is uneven.

10.2 Business Performance & Operations

Area	Inference
Lead generation	Organic search is the largest lead origin at about 2,300, followed by paid search (about 1,600), social (about 1,350) and unknown (about 1,100). Organic and paid search together supply roughly half of all leads, and the unknown origin is a visible attribution gap. Direct traffic, email and referral each contribute about 500 or fewer.
Closed deals	home_decor leads with roughly 105 closed deals, followed by health_beauty (about 93) and car_accessories (about 77). computers and pet are the smallest of the segments shown, at about 30–35 deals each.
Payment behaviour	credit_card dominates installment volume, at about 0.27M against about 0.02M for boleto and under 0.01M for voucher, debit_card and not_defined. The visual sums installments, so bar height reflects both payment volume and plan length.
Seller concentration	The top ten sellers each contribute roughly 0.16M–0.25M in sales, with a gradual decline from the first to the tenth. Sales are not dominated by a single seller among the leaders.
Customer satisfaction	Of about 99K reviews, roughly 58% are five-star and about 77% are four or five stars. One-star reviews (about 12%) outnumber two-star reviews (about 3%), a polarised J-shaped pattern which suggests dissatisfaction comes from clear service or product failures rather than mild disappointment.
Inventory vs demand	For the eight categories shown, average inventory level sits well above the average demand forecast in every case, indicating comfortable stock cover. Inventory data is self-generated and kept separate from Olist products, so this illustrates the reporting capability rather than an operational finding.

10.3 Insights and Business Implications

Stakeholder	Insight	Implication
VP, Sales & Revenue	Revenue tracks order volume and is concentrated in a few states.	Growth depends on order volume; regional concentration is a diversification question.
Regional Sales Manager	93.22% on-time rate with a small share of late deliveries.	Late deliveries are the lever for further improving fulfilment.
CRM / Customer Insights Lead	Reviews are strongly positive with a distinct cluster of one-star reviews.	Investigating one-star orders against delivery outcomes can target dissatisfaction.
Merchandising / Product Lead	Revenue is spread across several leading categories.	No single-category dependency; health and beauty, gifts and home goods lead.
Dealer / Partner Manager	Top sellers contribute at similar levels.	Marketplace revenue is not reliant on one seller.
Inventory Planner	Inventory exceeds forecast demand in the categories shown.	The reporting supports stock-versus-demand monitoring once a validated product mapping exists.
Marketing Analyst	Search drives about half of leads; a notable share has unknown origin.	Improving origin capture would sharpen channel attribution.

10.4 Reading Notes

- Monthly views aggregate all years in the selected date range, so months covered by fewer years carry lower totals. The September–December decline should be read together with the coverage of the underlying order dates rather than as a standalone seasonal conclusion.
- The Order Status Distribution counts order records at order-item grain (112.65K), while the Total Orders card reports distinct orders (99K); the two figures differ for that reason.
- The installments visual sums payment_installments, so it measures total installment volume rather than the typical number of installments per payment.
- Inventory and demand forecast figures are self-generated and are not linked to Olist product IDs.

11. Testing, Validation and Evidence

Step	Test Performed	Evidence Retained
1	Source inputs confirmed.	Source inventory and file/table checks.
2	Ingestion and staging jobs run.	Successful job logs and staging counts.
3	Profiling and cleaning run.	Reports, rules and output checks.
4	Warehouse loaded.	Fact/dimension counts and schema checks.
5	Validation SQL run.	Report with measured results.
6	Metadata and lineage verified.	Repository queries, mapping and diagram.
7	Power BI refreshed.	Both pages and sample SQL reconciliation.
8	Release frozen.	Git commit/tag and deliverables inventory.

11.1 Evidence Retained

- Full Pentaho job success screenshot.
- PostgreSQL staging and warehouse row-count evidence.
- Latest validation report with exceptions explained.
- Metadata repository query results and final lineage diagram.
- Power BI screenshots/export of both pages.
- Git repository location and final commit/tag.

Validation figures were measured from the live database, and reconciliation results and explained exceptions are held in the validation report and workbook.

12. Outcomes, Limitations and Improvements

12.1 Outcomes

- Multi-format ingestion through Pentaho and PostgreSQL staging.
- Documented profiling, quality rules and cleansing between staging and the warehouse.
- Warehouse model for sales, payments, reviews, inventory, marketing and closed deals.
- Governance artifacts for mapping, lineage, metadata and validation.
- Two-page Power BI report for executive and operational analysis.
- A release frozen through a tagged Git commit with a deliverables inventory.

12.2 Limitations

- Inventory IDs are not assumed to match Olist product IDs, so inventory is not joined to dim_product.
- Payment and review facts do not use the sales date slicer in the current model.
- Order-item counts and distinct order counts are reported separately.
- Inventory and demand forecast figures are self-generated rather than drawn from an operational system.

12.3 Future Improvements

Automate regression checks; create a validated inventory-to-product mapping if needed; improve tooltips and seller labels while retaining traceability; capture lead origin more completely to reduce unknown attribution; and consider incremental loading and scheduled refresh if the platform is extended.

13. Final Deliverables Checklist

#	Deliverable	Final Status
1	Charter, BRD, source inventory and architecture	Confirmed — current and consistent with scope.
2	Source conversions and Pentaho transformations/jobs	Confirmed — executed successfully with correct targets.
3	Staging schema and data dictionary	Confirmed — matches actual staging tables.
4	Profiling reports	Confirmed — contain counts, fields, types, missing values, duplicates and unique values.
5	Cleaning rules and quality comparison	Confirmed — applied rules and measured effects documented.
6	Warehouse schema and data dictionary	Confirmed — matches live schema and relationships.
7	Source-to-target mapping	Confirmed — matches final ETL and target columns.
8	Metadata SQL and run guide	Confirmed — executed and verified.
9	End-to-end and column-level lineage	Confirmed — matches final flow.
10	Validation SQL/report	Confirmed — rerun after final load; exceptions explained.
11	Orchestration/audit evidence	Confirmed — shows full workflow completion.
12	Power BI report and evidence	Confirmed — both pages readable and checked against SQL.
13	README and repository structure	Confirmed — setup, execution and file locations documented.
14	Final report and presentation	Confirmed — consistent with implementation and results.
15	Git release/tag and backup	Confirmed — final files versioned and archived.

Conclusion

ESADP demonstrates an end-to-end analytics workflow from heterogeneous source data through ingestion, staging, quality processing, a PostgreSQL warehouse, governance documentation and Power BI reporting. Execution evidence and validation results accompany this submission so that a reviewer can trace every reported measure back to its warehouse source and original file.

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Appendix A. Glossary

Term	Meaning
Medallion architecture	A layered data design with Bronze (raw), Silver (cleansed) and Gold (analytics-ready) zones.
Bronze zone	Raw ingestion and staging; data lands as received from each source.
Silver zone	Cleansing, validation, standardization and business-rule transformation.
Gold zone	Curated star-schema warehouse and the analytics layer.
Star schema	A warehouse model of central fact tables related to surrounding dimension tables.
Fact table	A table of measurable business events, such as order items, payments or reviews.
Dimension table	A table of descriptive attributes, such as date, customer, product or seller.
Grain	The level of detail represented by one row of a fact table.
Pentaho (Spoon)	The data integration tool used to build ETL transformations (.ktr) and jobs (.kjb).
Staging table	A raw landing table in PostgreSQL holding source data before cleansing.
MQL	Marketing-qualified lead.
Average Order Value	Total sales revenue divided by the number of distinct orders.
On-Time Delivery Rate	Share of delivered orders that arrived on or before the estimated date.
Data lineage	The traceable path of a data element from source through staging and warehouse to the dashboard.
Source-to-target mapping	A field-level record of how each source column maps to a target column and its transformation rule.
Crosswalk	A validated mapping between two sets of identifiers, such as inventory and Olist product IDs.